

Oct 13, 2024 (Due: 08:00 Oct 20, 2025)

1. Let $A \in \mathbb{C}^{m \times n}$, $b \in \mathbb{C}^m$. Show that solving the least squares problem

$$\min_{x \in \mathbb{C}^n} \|b - Ax\|_2$$

is mathematically equivalent to solving the augmented system

$$\begin{bmatrix} I_m & A \\ A^* & 0 \end{bmatrix} \begin{bmatrix} r \\ x \end{bmatrix} = \begin{bmatrix} b \\ 0 \end{bmatrix}.$$

2. Let $A \in \mathbb{C}^{m \times n}$ and $X \in \mathbb{C}^{n \times m}$. Suppose that for any $b \in \mathbb{C}^m$, $x = Xb$ is always a minimizer of the least squares problem $\min_x \|Ax - b\|_2$. Show that $AXA = A$ and $(AX)^* = AX$.

3. Given two matrices $A \in \mathbb{C}^{m \times n}$ and $B \in \mathbb{C}^{p \times n}$ with $p \leq n \leq m + p$. Show that there exist unitary matrices $Q \in \mathbb{C}^{n \times n}$, $Z \in \mathbb{C}^{m \times m}$ and upper triangular matrices $R \in \mathbb{C}^{p \times p}$, $T \in \mathbb{C}^{n \times n}$ such that

$$A = Z \begin{bmatrix} T \\ 0 \end{bmatrix} Q, \quad B = \begin{bmatrix} 0 & R \end{bmatrix} Q.$$

4. Write a program to compute the QR factorization of a general complex matrix $A \in \mathbb{C}^{m \times n}$ with $m \geq n$ using CGS and MGS, with and without reorthogonalization. Visualize the *componentwise* loss of orthogonality $|Q^*Q - I_n|$ and the normwise residual $\|A - QR\|_F$ with a few examples.

5. Find the “best” straight line that approximately passes through the data set $\{(n, \ln n) \in \mathbb{R}^2 : n \in \{3, 4, \dots, 10\}\}$. Visualize your result and clarify in what sense your solution is the best.

6. (H) Implement the Arnoldi process based on Householder-QR.

7. (optional) Let $A \in \mathbb{R}^{m \times n}$ with full column rank. Establish the connection between the Householder-QR algorithm applied to the matrix $[0, A^\top]^\top \in \mathbb{R}^{(m+n) \times n}$ and the MGS algorithm applied to A .